

EDS Theory Assignment 2

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Problem Statement:

A dataset (House_Price_Prediction_Dataset.csv) containing house listing records has been provided. Each record contains features such as Area (sq ft), Bedrooms, Bathrooms, Floors, YearBuilt, Location, Condition, Garage (Yes/No), and Price. You are required to analyze this dataset using NumPy for numerical computations and Pandas for data manipulation. Based on the dataset, formulate and solve 20 problem statements (10 NumPy + 10 Pandas).

Complete Code:

```
import numpy as np
import pandas as pd
np.set_printoptions(suppress=True)

# Load Dataset
df = pd.read_csv('House_Price_Prediction_Dataset.csv')

# Convert to NumPy arrays
prices = df['Price'].to_numpy()
areas = df['Area'].to_numpy()
bedrooms = df['Bedrooms'].to_numpy()
garages = df['Garage'].to_numpy()

# NUMPY OPERATIONS

# 1. First 10 house prices
print('1. First 10 House Prices:')
print(prices[:10])

# 2. Total sum of all prices
print('2. Total Sum of All Prices:')
print(np.sum(prices))

# 3. Average house price
print('3. Average House Price:')
print(round(np.mean(prices), 2))

# 4. Maximum house price
print('4. Maximum House Price:')
print(np.max(prices))
```

```

# 5. Minimum house price
print('5. Minimum House Price:')
print(np.min(prices))

# 6. Median area across all houses
print('6. Median Area (sq ft):')
print(np.median(areas))

# 7. Standard deviation of house prices
print('7. Std Dev of House Prices:')
print(round(np.std(prices), 4))

# 8. Number of houses with Garage
print('8. Number of Houses With Garage:')
print(np.sum(garages == 'Yes'))

# 9. Top 10 prices in descending order
print('9. Top 10 Prices (Descending):')
print(np.sort(prices)[::-1][:10])

# 10. Index of the most expensive house
print('10. Index of Most Expensive House:')
print(np.argmax(prices))

```

PANDAS OPERATIONS

```

# 11. First 5 rows
print('11. First 5 rows:')
print(df[['Area', 'Bedrooms', 'Bathrooms', 'Price']].head().to_string(index=False))

# 12. Last 5 rows
print('12. Last 5 rows:')
print(df[['Area', 'Bedrooms', 'Bathrooms', 'Price']].tail().to_string(index=False))

# 13. Column names and data types
print('13. Column Info:')
print(df.dtypes.to_string())

# 14. Shape of the dataset
print('14. Dataset Shape:')
print(df.shape)

# 15. Average price per Location
print('15. Avg Price by Location:')
print(df.groupby('Location')['Price'].mean().round(2).to_string())

```

```

# 16. Count of houses per Location
print('16. Houses per Location:')
print(df['Location'].value_counts().to_string())

# 17. Average area per Location
print('17. Avg Area by Location:')
print(df.groupby('Location')['Area'].mean().round(2).to_string())

# 18. Top 5 most expensive houses
print('18. Top 5 Most Expensive Houses:')
print(df.nlargest(5, 'Price')[['Location', 'Area', 'Bedrooms', 'Price']].to_string(index=False))

# 19. Bottom 5 cheapest houses
print('19. Bottom 5 Cheapest Houses:')
print(df.nsmallest(5, 'Price')[['Location', 'Area', 'Bedrooms', 'Price']].to_string(index=False))

# 20. Summary statistics
print('20. Summary Statistics:')
print(df[['Area', 'Bedrooms', 'Bathrooms', 'Floors', 'YearBuilt', 'Price']].describe().round(2).to_string())

```

NumPy Operations

1. Display the first 10 house prices as a NumPy array.

Code:

```
print(prices[:10])
```

Output:

```

1. First 10 House Prices:
[149919 424998 266746 244020 636056 93262 448722 594893 652878 340375]

```

2. Find the total sum of all house prices.

Code:

```
print(np.sum(prices))
```

Output:

```

2. Total Sum of All Prices:
1075353710

```

3. Compute the average house price across all records.

Code:

```
print(round(np.mean(prices), 2))
```

Output:

```

3. Average House Price:
537676.86

```

4. Identify the maximum house price in the dataset.

Code:

```
print(np.max(prices))
```

Output:

```
4. Maximum House Price:
999656
```

5. Identify the minimum house price in the dataset.

Code:

```
print(np.min(prices))
```

Output:

```
5. Minimum House Price:
50005
```

6. Calculate the median area (sq ft) across all houses.

Code:

```
print(np.median(areas))
```

Output:

```
6. Median Area (sq ft):
2833.0
```

7. Compute the standard deviation of house prices.

Code:

```
print(round(np.std(prices), 4))
```

Output:

```
7. Std Dev of House Prices:
276359.7299
```

8. Find the total number of houses that have a Garage (Garage = 'Yes').

Code:

```
print(np.sum(garages == 'Yes'))
```

Output:

```
8. Number of Houses With Garage:
962
```

9. Sort all house prices in descending order and show the top 10.

Code:

```
print(np.sort(prices)[::-1][:10])
```

Output:

```
9. Top 10 Prices (Descending):
[999656 999453 998128 998084 997719 997226 997176 996931 996740 996357]
```

10. Find the index (position) of the most expensive house in the dataset.

Code:

```
print(np.argmax(prices))
```

Output:

```
10. Index of Most Expensive House:  
1004
```

Pandas Operations

11. Display the first 5 rows of the dataset.

Code:

```
print(df[['Area', 'Bedrooms', 'Bathrooms', 'Price']].head().to_string(index=False))
```

Output:

11. First 5 rows:

Area	Bedrooms	Bathrooms	Price
1360	5	4	149919
4272	5	4	424998
3592	2	2	266746
966	4	2	244020
4926	1	4	636056

12. Display the last 5 rows of the dataset.

Code:

```
print(df[['Area', 'Bedrooms', 'Bathrooms', 'Price']].tail().to_string(index=False))
```

Output:

12. Last 5 rows:

Area	Bedrooms	Bathrooms	Price
4994	5	4	295620
3046	5	2	580929
1062	5	1	476925
4062	3	1	161119
2989	5	1	482525

13. Show the column names and data types of the dataset.

Code:

```
print(df.dtypes.to_string())
```

Output:

13. Column Names and Data Types:

Id	int64
Area	int64
Bedrooms	int64

Bathrooms	int64
Floors	int64
YearBuilt	int64
Location	object
Condition	object
Garage	object
Price	int64

14. Find the total number of rows and columns in the dataset.

Code:

```
print(df.shape)
```

Output:

```
14. Dataset Shape (rows, columns):  
(2000, 10)
```

15. Calculate the average house price per Location.

Code:

```
print(df.groupby('Location')['Price'].mean().round(2).to_string())
```

Output:

```
15. Avg Price by Location:  
Location  
Downtown    536059.66  
Rural        538613.94  
Suburban    557416.33  
Urban       518963.55
```

16. Find the total number of houses per Location.

Code:

```
print(df['Location'].value_counts().to_string())
```

Output:

```
16. Houses per Location:  
Location  
Downtown    558  
Urban       485  
Suburban    483  
Rural       474
```

17. Find the average area (sq ft) per Location.

Code:

```
print(df.groupby('Location')['Area'].mean().round(2).to_string())
```

Output:

```
17. Avg Area by Location:  
Location  
Downtown    2813.63
```

Rural	2745.95
Suburban	2794.02
Urban	2786.23

18. Identify the top 5 houses with the highest price.

Code:

```
print(df.nlargest(5, 'Price')[['Location', 'Area', 'Bedrooms', 'Price']].to_string(index=False))
```

Output:

```
18. Top 5 Most Expensive Houses:
Location  Area  Bedrooms  Price
Suburban  3099         1  999656
Urban     566         4  999453
Downtown  736         3  998128
Downtown  2860        2  998084
Urban     2985         4  997719
```

19. Identify the bottom 5 houses with the lowest price.

Code:

```
print(df.nsmallest(5, 'Price')[['Location', 'Area', 'Bedrooms', 'Price']].to_string(index=False))
```

Output:

```
19. Bottom 5 Cheapest Houses:
Location  Area  Bedrooms  Price
Rural     4260         2  50005
Suburban  2871         4  50064
Urban     509         4  51082
Downtown  4368         5  51845
Urban     4138         3  52024
```

20. Display summary statistics for key numeric columns.

Code:

```
print(df[['Area', 'Bedrooms', 'Bathrooms', 'Floors', 'YearBuilt', 'Price']].describe().round(2).to_string())
```

Output:

```
20. Summary Statistics:
      count      Area  Bedrooms  Bathrooms  Floors  YearBuilt      Price
count  2000.00    2000.00    2000.00    2000.00  2000.00    2000.00    2000.00
mean   2786.21         3.00         2.55     1.99    1961.45   537676.86
std    1295.15         1.42         1.11     0.81     35.93   276428.85
min     501.00         1.00         1.00     1.00    1900.00    50005.00
25%    1653.00         2.00         2.00     1.00    1930.00   300098.00
50%    2833.00         3.00         3.00     2.00    1961.00   539254.00
75%    3887.50         4.00         4.00     3.00    1993.00   780086.00
max    4999.00         5.00         4.00     3.00    2023.00   999656.00
```

Thank you